QUESTION 1

Which of the following are key aspects of efficiency in embedded systems?

Answer is Correct **Score: 1 / 1**

Options

Correct

- [x] A. Power efficiency

- [ ] B. Compilation speed

Correct

- [x] C. Resource efficiency

Correct

- [x] D. Code maintainability

FEEDBACK

Solution: Efficiency in embedded systems includes optimal use of CPU cycles, memory, power, and writing maintainable code.

Link: https://youtu.be/DQ2cnYWQHD8?feature=shared&t=29

QUESTION 2

Why is efficiency especially critical in systems like elevators or lifts?

Answer is Correct **Score: 1 / 1**

Options

A. They require fast internet

B. They are restarted frequently

Correct

C. They are expected to run continuously without failure

D. They only operate in daytime

FEEDBACK

Solution: Systems like elevators are expected to run 24x7 and respond instantly, making efficiency in memory and performance crucial.

Link: https://youtu.be/DQ2cnYWQHD8?feature=shared&t=251

QUESTION 3

What is the risk of storing 8-bit values in 32-bit integers in embedded systems?

Answer is Correct **Score: 1 / 1**

Options

A. It increases power usage slightly

B. It saves execution time

Correct

C. It wastes memory unnecessarily

D. It speeds up floating-point math

FEEDBACK

Solution: Using a 32-bit variable for 8-bit data wastes space, especially problematic when dealing with arrays or limited memory devices. Link: https://youtu.be/DQ2cnYWQHD8?feature=shared&t=399

QUESTION 4

Which strategies help minimize global state in embedded C programs?

Not Attempted **Score: 0 / 1**

Options

- [ ] A. Avoid local function parameters
- [ ] B. Use dynamic memory allocation
- [ ] C. Use local variables instead of globals — Expected
- [ ] D. Minimize dynamic allocation — Expected

FEEDBACK

Solution: Avoiding global state and minimizing dynamic memory help improve modularity, reduce bugs, and conserve memory in embedded code.
Link: https://youtu.be/6KVgKWnGqJA?feature=shared&t=541

QUESTION 5

What is the main advantage of avoiding floating-point calculations in embedded systems?

Answer is Wrong **Score: 0 / 1**

Options

Wrong

A. It improves accuracy

Expected

B. It uses fewer CPU cycles

C. It increases memory usage

D. It is more human-readable

FEEDBACK

Solution: Floating-point operations are slow and may require software emulation; using fixed-point arithmetic saves clock cycles. Link: https://youtu.be/DQ2cnYWQHD8?feature=shared&t=1036